Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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Experiment 7

Bagging, Boosting, and Stacked Ensemble Models

Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalisation.
- To analyse the effect of ensemble methods on bias and variance.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset (loaded from `sklearn.datasets`)

- Total samples: 569 (357 Benign, 212 Malignant)
- Features: 30 numerical attributes
- Train/Test split: 80%/20% with StandardScaler applied

Theory

Bagging (Bootstrap Aggregation)

Trains multiple models on different bootstrap samples and aggregates predictions. Reduces variance; each model is trained independently.

Boosting (AdaBoost / Gradient Boosting)

Models are trained *sequentially*; each new model corrects errors of previous ones. Reduces bias; emphasises misclassified samples.

Stacked Ensemble (Stacking)

Combines heterogeneous base models (SVM, Naive Bayes, Decision Tree) and trains a meta-learner (Logistic Regression) on their out-of-fold predictions. Often achieves superior performance by leveraging diverse strengths.

Steps for Implementation

1. Load Breast Cancer dataset; apply StandardScaler.
2. 80–20 train/test split (stratified).
3. BaggingClassifier: 5-fold CV over `n_estimators`, `max_samples`.
4. AdaBoostClassifier: 5-fold CV over `n_estimators`, `learning_rate`.
5. GradientBoostingClassifier: 5-fold CV over `n_estimators`, `learning_rate`, `max_depth`.
6. StackingClassifier (SVM + Naive Bayes + Decision Tree → Logistic Regression): 5-fold CV.
7. Evaluate all models on the held-out test set.

Hyperparameter Evaluation Results

Bagging Results

Table 1: Bagging – 5-Fold CV (base: DecisionTree)

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.8	94.73	0.9571
50	0.8	94.95	0.9593
100	**1.0**	**95.16**	**0.9609**

Boosting Results

Table 2: AdaBoost / GradientBoosting – 5-Fold CV

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
AdaBoost			
50	0.5	96.70	0.9737
100	**1.0**	**98.02**	**0.9844**
200	0.1	97.14	0.9773
Gradient Boosting			
100	0.1	95.16	0.9614
200	**0.05**	**96.04**	**0.9684**
100	0.1	93.41	0.9470

Stacked Ensemble Results

Table 3: Stacking – 5-Fold CV

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM + Naive Bayes + Decision Tree	Logistic Regression	95.43	0.9664

Performance Comparison (Test Set)

Table 4: Final Test-Set Performance (114 samples)

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC-ROC**
Bagging (DT, 100)	93.86%	0.9452	0.9583	0.9517	0.9927
AdaBoost (100, lr=1.0)	95.61%	0.9467	0.9861	0.9660	0.9818
GradBoost (200, lr=0.05)	95.61%	0.9467	0.9861	0.9660	0.9911
Stacking (SVM+NB+DT)	**97.37%**	**0.9726**	**0.9861**	**0.9793**	**0.9937**

Observation Questions

- **How does Bagging reduce variance?** By training 100 trees independently on random bootstrap samples (80–100% of data), random fluctuations are averaged out, reducing the overall prediction variance compared to a single decision tree.
- **How does Boosting address model bias?** Each successive model focuses on samples misclassified by previous models. AdaBoost re-weights samples, while Gradient Boosting fits the residual error, progressively reducing bias (AdaBoost CV accuracy reached 98%).
- **Why does stacking benefit from heterogeneous models?** SVM, Naive Bayes, and Decision Tree each capture different patterns. The Logistic Regression meta-learner learns the optimal combination, correcting weaknesses of each individual model.
- **Which ensemble method performed best and why?** Stacking (97.37% test accuracy) outperformed all others by combining diverse base classifiers. AdaBoost was the best single-algorithm boosting method (98% CV accuracy), but its test accuracy matched Gradient Boosting at 95.61%.

Conclusion

Stacking achieved the highest test accuracy (97.37%) and AUC (0.9937), demonstrating that combining diverse base learners with a meta-learner is the most powerful ensemble strategy for this dataset. AdaBoost showed the highest cross-validation accuracy (98%) among single-algorithm methods, confirming that sequential boosting effectively reduces model bias.

References

- Scikit-learn: Ensemble Methods

- Bagging Classifier
- UCI Dataset